

Jia-Rong Cheng

Tainan, Taiwan

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PlusRon

Education

National Central University | Master degree - CSIE

Zhongli, Taiwan

M.S. IN COMPUTER SCIENCE AND INFORMATION ENGINEERING

GPA : 4.3 / 4.3

September 2023 - July 2025

National Yunlin University of Science and Technology | Bachelor degree - CSIE

Yunlin, Taiwan

B.S. IN COMPUTER SCIENCE AND INFORMATION ENGINEERING

GPA : 3.98 / 4.0

September 2019 - June 2023

Firmware & OS Internals Projects

Bare-metal ARM Cortex-M0 | SSD Controller Simulator

PlusRon/Firmware-STM32-SSD-Emulator

FIRMWARE DEVELOPER

March 2026 - Present

• Bootstrapping & HAL Design

- Standalone GNU Toolchain built with Arm-GCC, Make, and OpenOCD for a CLI-based environment.
- Linker Scripts and Startup Code authored to manage memory layout, Vector Table configuration, and SCB-based auto-recovery.
- Custom HAL (STM32F072) engineered for precise peripheral control.

• Interrupt-Driven Architecture

- DMA and Interrupt Service Routines (ISR) integrated with Ring Buffers to achieve high-throughput, non-blocking I/O.
- UART IDLE Detection and SysTick interrupts utilized for asynchronous event handling and system timing.

• FTL (Flash Translation Layer) Management

- L2P Mapping and Out-of-place Updates implemented to manage LBA-to-PBA address translation and flash media logic.
- Linked-list-based Garbage Collection (GC) developed for efficient page-pooling and stale data reclamation.

• Communication Protocol & Reliability

- NVMe-like Protocol Parser engineered with Checksum validation and Endianness handling.
- Flow Control (RTS/CTS) with ORE (Overrun Error) recovery to prevent data overflows and race conditions.
- Python-based Unit Testing developed to automate host-to-device protocol validation and command verification.

Linux Kernel | Custom Syscall for Page Table Traversal

PlusRon/linux_kernel

OS DEVELOPER

December 2025

- Custom System Call implemented for manual PGD-to-PTE traversal to retrieve physical addresses from virtual memory mappings of target processes.
- User-Kernel Data Exchange engineered to export memory mappings metadata for user-space diagnostics and dmesg verification.

Master's Thesis

SLC-E : Segment-Level Clustering and Extraction based Cross-Modal Video Retrieval and its Applications in Video Captioning

PlusRon/SLC-E-base-VC-DST

AI DEVELOPER

June 2025

• Video Retrieval (SLC-E)

- Integrated ViT with K-medoids++ (SLC) for feature compression and Cross-Modal Attention (CMSE) for segment alignment.
- Leveraged Contrastive Learning for model training and performance optimization.

• Video Captioning (DST)

- Dual-Stream Transformer implements a coupled architecture.
- Utilizing Auto-regressive decoding and Teacher Forcing to optimize generative quality.

- **Benchmark Evaluation**
 - Achieved 6.4% gain in BLEU-4 and 4.8% in R@1 on MSR-VTT.
 - Confirming high-precision semantic alignment across Retrieval and Captioning.

Work Experience

NCKU Tainan Industrial Senior High School | Intern Teacher

 **PlusRon/NCKU-ATIHS-intern**

DEPT. OF INFORMATION AND ELECTRONIC ENGINEERING

January 2026 - Present

- **Hardware Instrumentation & Debugging**
 - Proficient in Oscilloscopes, DMMs, and Signal Generators.
 - OPA / MOS / BJT / Digital Logic breadboard assemblies and perform systematic troubleshooting.
- **Circuit Simulation App**  **Flutter_app-Electronics_laboratory_project**
 - Developed a Flutter PWA based on Wasm and Skia, featuring CI/CD-integrated Cache Busting and asynchronous Service Worker monitoring.
 - Simulates real-time circuit hardware behavior through high-performance software logic to enable interactive instructional demonstrations.

Skills

Programming Languages	C (Embedded) , Python (Testing/AI) , C++ , Shell Script , Dart
Embedded Systems	STM32 (Cortex-M0) , DMM , Oscilloscope , Electronics , Linker Scripts
OS Internals	Bootstrapping , Syscall , Virtual Memory , Page Table Traversal (PGD/PTE)
AI & Machine Learning	PyTorch , Transformers , Vision Transformers , Contrastive Learning
Application & Web	Flutter , PWA , Responsive UI , Cache-Busting , Service Worker
Toolchains & DevOps	Arm-GCC , GNU Make , OpenOCD , GDB , CI/CD , Linux CLI , Git